Final Project: Pick a Project

Computer Science - Week 10

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Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

Role	Name	Email
Developer (You)	Audrey Houghton	audreymh@udel.edu
Friend (optional)	Jhayce Hawley	jhayce@udel.edu

Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea and a paragraph describing how that paper supports your design concept.

        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider creating an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the ChemPy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages from uploaded images.
    - **Mechanical Engineering** majors may consider a project that somehow uses both the NumPy and SciPy Python libraries.

My idea is to create a website where a novice biochemical researcher can search up the procedure they intend to do and get detailed instructions with images about how to do it. They can also input numbers in relation to how much data they want and get instructions that include quantities of materials they should use.

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

As far as I can tell, there are no interactive tools for explaining and depicting biotechnology lab procedures. However, there are multiple guides to laboratory procedures such as this one made by an AP Biology teacher: https://www.biointeractive.org/sites/default/files/Biotechnology_Guide.pdf and this one by BIORAD: https://www.bio-rad.com/webroot/web/pdf/lse/literature/4006101.pdf. There is also this webpage by Amgen, which includes visuals of lab procedures: https://www.amgenbiotechexperience.com/curriculum-resources/abe-visual-guides.

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

The purpose of this site would be to assist novice biochemical researchers with designing and understanding experimental procedures so that they don't have to spend the time reading about it in a bunch of different articles. They can get all the information they need about the procedure and what they need to do in one place.

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

The goal of this program is to take input about what procedure the user wants to perform and the number of samples they want in each trial and to return a page that explains the procedure with images and quantities found from processing the number of samples input by the user.

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

The user will need to input the procedure that they want to perform (they will choose from a list) and the number of samples they want in each trial of the experiment.

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

The computer will use the requested procedure to inform which step-by-step procedure and images to show. The computer will use the desired number of samples to calculate the quantities of different materials that are needed.

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

The user will get back a numbered list description of the procedure they should perform,

complete with images that depict the procedure and containing accurate quantities for what they should use.

Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things" (nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

Procedure, materials

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

Details I need: What procedure the user wants to perform, Number of samples the user wants per treatment group (ex. For immunofluorescence staining and imaging, the final number of cell data points they want per treatment group), What chemicals will be used
Details I will ignore: Where the chemicals come from, Timeframe the user has to complete the experiment

Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

Understanding the laboratory procedures well enough to write out in-depth procedures, accounting for different objectives/use of different chemicals within one procedure, calculating all the quantities needed of different chemicals just given the number of data points the user wants, adding images created outside of Python into the website.

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow

learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

No LLM
Work times: Tuesdays 8:30-10:30pm, Thursdays 5-7pm

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

Create something that works, is pretty close to scientifically accurate, and could be somewhat useful to novice researchers; expand my coding capabilities and thoroughly use everything I've learned in this class; create something impressive enough to put on a resume

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

Hello AI friend Jhayce! I'm supposed to ask you to help me brainstorm. My idea is to make a website that has multiple options for lab procedures that the user (a novice researcher) wants to perform. When they pick one, they can input their specific goal/what chemicals they plan to use and the number of data points they want at the end. The website would then return a detailed, step-by-step procedure with images and quantities of chemicals calculated based off of the final number of data points the user wants. What are three features this website should have, and apparently, what solutions already exist?

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

AI friend Jhayce says my idea seems useful but challenging to implement.
Feature suggestions:

1. *Clean user interface, easy to navigate*
2. *Minimize technical jargon*
3. *High detail in each step*
4. *Use a web-scraper to find step-by-step procedures online for many lab processes and input directly into the program*

This idea is similar to traditional lab manuals but a modern take that is interactive with the user.

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

AI friend Jhayce, I don't know how to use a web-scraper and neither do you. Maybe that's too complicated for an introductory programming project. Howsabout a simpler alternative?? Oh, I could just manually find lab procedures online and copy them? Wow, that sounds a lot simpler and much more appropriate for an introductory programming project! Thank you AI friend Jhayce!

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

This conversation hasn't really changed my project idea, but it gave me the idea to copy some of the procedures from existing lab manuals. Thinking more about the scope of the project is also making me wonder if I should just focus on one laboratory procedure.

Part 6: Resource Planning

1. What resources will you use if you get stuck?

I will use my friends (particularly Jhayce), Nazim, and the CISC106 Discord.

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

I do not plan on using an LLM.

Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

I did not use an LLM to assist with this assignment.

Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.